

SANTHANAKRISHNAN M

SENIOR FRONTEND ENGINEER | UI ARCHITECT | GENAI & FULL-STACK UI PROFESSIONAL

+917708923153

santhanakrishnanstark@gmail.com

LinkedIn

GitHub

Front-end Blog

Senior Front-end Engineer & UI Architect experience specializing in enterprise-scale UI engineering, modern React.js ecosystems & scalable front-end architecture, targeting senior leadership opportunities in Front-end Engineering, UI Architecture across diverse industries.

CORE COMPETENCIES

Frontend Architecture & UI Engineering
Design Systems & Component Libraries
Generative AI & LLM Integration
Retrieval Augmented Generation (RAG)
Microservices & API Integration
Full Stack Web Development
Cloud-Native Application Development
CI/CD Pipeline Automation
Performance Optimization
State Management & UI Patterns
Agile Product Engineering

TECHNICAL SKILLS

Front-end Technologies: React.js, React Hooks, Redux Toolkit, Next.js, TypeScript, Vue.js, JavaScript (ES6+), Context API, React Router, Web Components, Atomic Design Systems, WCAG 2.1 Accessibility, Responsive Design
Design Systems & UI Libraries: MUI, MUI X (DataGrid, Pro), Tailwind CSS, Styled Components, High-Performance Reusable Component Libraries, Custom UI Design Patterns
GenAI, LLM & RAG Stack: LangChain, LangGraph, Prompt Engineering, Agentic AI, Model Context Protocol (MCP), Function Calling, Tool Use, LLM Token Streaming, Server-Sent Events (SSE), Multimodal AI
LLM Providers & Vector Search: OpenAI (GPT-4), Anthropic (Claude), Google Gemini, Hugging Face Transformers, Ollama, Vertex AI, Chunking, Embeddings, Semantic Search, Hybrid Search, Chroma DB, pgvector, Pinecone, Re-ranking

PROFILE SUMMARY

- Senior Frontend Engineer with nearly 7 years of experience in building scalable, enterprise-grade web applications using **React.js, TypeScript, Vue.js, and modern JavaScript frameworks**, with strong expertise in UI architecture and component-driven design systems.
- Experience in leading **frontend architecture modernization**, including migration of legacy systems to **framework-agnostic Web Components**, improving scalability, reusability & reducing time-to-market.
- Expertise in integrating **Generative AI, LLMs & RAG-based systems** into production applications using tools like **LangChain, LangGraph, SSE streaming & vector databases**, enabling real-time AI-driven UX.
- Skilled in **Python, FastAPI, Node.js, REST/GraphQL APIs & Microservices integration**, ensuring seamless frontend-backend orchestration in complex distributed systems.
- Showcased leadership in driving **performance engineering, CI/CD automation, testing & DevOps practices**, achieving high test coverage, improved release velocity & reduced production defects.
- Proven skills to lead and mentor engineering teams while collaborating with **product managers, ML engineers, and cross-functional stakeholders** to deliver high-impact AI-enabled digital products.

NOTABLE ACCOMPLISHMENTS

- Led the end-to-end design and development of a **DevOps internal web application** that replaced manual GitHub-based workflows with a centralized dashboard, significantly improving deployment visibility and reducing operational effort for engineering teams.
- Designed **full-scale frontend application from scratch**, defining UI architecture, component structure, and technical standards, ensuring scalability, maintainability, and design system adoption.
- Successfully mentored & guided **junior frontend engineers while simultaneously handling multiple projects**, improving team productivity & ensuring high-quality deliverables across parallel initiatives.
- Spearheaded modernization of frontend systems by building reusable and scalable UI components, contributing to improved development efficiency and reduced duplication across applications.
- Recognized for **consistent high performance and delivery excellence**, contributing to multiple successful project releases while maintaining strong code quality and architectural standards.

WORK EXPERIENCE

Infosys Ltd, Bangalore | Specialist Programmer (Senior Frontend Engineer & UI Lead) | May'22 – Present

- Led end-to-end UI engineering initiatives and frontend standards implementation leveraging React Hooks, MUI, and MUI X (DataGrid, AgGrid), enabling creation of reusable component libraries and scalable UI design systems.
- Directed modernization efforts by transforming legacy, framework-dependent modules into framework-agnostic Web Components, improving cross-product portability by **30%** and reducing feature time-to-market by **25%**.
- Developed advanced frontend integrations for GenAI and RAG-based APIs, enabling real-time token streaming via Server-Sent Events (SSE), optimistic UI rendering, multi-source citation display, and resilient error recovery, reducing perceived chat latency by **40%**.
- Managed frontend architecture for REST and GraphQL API consumption, including Microservice integration patterns, JWT-based role authorization, and structured UI error-handling mechanisms.
- Implemented UI observability frameworks using Azure Application Insights to monitor LLM-driven user interactions, token consumption, latency patterns, and failure scenarios, identifying and preventing **12+ production issues**.
- Strengthened test coverage strategy by achieving **90%+ test coverage** using Vitest and Playwright, reducing production regressions by **50%** and post-release hotfixes by **35%**.
- Optimized CI/CD workflows through GitHub Actions automation, enforced SonarQube quality gates, and containerized enterprise deployments using Docker, improving release velocity by **30%**.
- Collaborated with ML engineers, data scientists, product managers, and business stakeholders to translate ML models into **6+ production-grade AI-driven customer-facing features**.
- Mentored and guided a team of **5+ frontend engineers** through code reviews, architecture discussions, and UI/API standards enforcement to ensure engineering consistency and quality.

Backend, APIs & Databases:

Python, FastAPI, Node.js, Express, REST APIs, GraphQL, Microservices, WebSockets, Event-Driven Architecture, PostgreSQL, MongoDB, Supabase, JWT, OAuth 2.0 Authentication

Testing, Quality & DevOps:

Vitest, Jest, React Testing Library, Playwright, Cypress, SonarQube Quality Gates, AWS (Lambda, API Gateway, S3, CloudWatch), Azure Application Insights, Docker, GitHub Actions CI/CD

EDUCATION

- **Master of Computer Applications (MCA), 2019**
Madras University, Chennai, India
- **Bachelor of Computer Applications (BCA), 2017**
Thiruvalluvar University, Cuddalore, India

CERTIFICATIONS

- Develop Advanced Enterprise Search and Conversation Applications — Google Cloud
- Integrate Vertex AI Search and Conversation into Voice and Chat Apps — Google Cloud
- Build Search & Recommendations Applications with AI Applications — Google Cloud
- AWS Certified Developer Associate — Infosys Accredited Certification
- React Professional & Python Programmer Core Accreditations — Infosys
- Back End Development and APIs Professional Certification — freeCodeCamp

- ② Led the development of **fully WCAG 2.1 compliant enterprise web application**, ensuring accessibility adherence across all custom UI components while collaborating with Apple client engineering teams.
- ② Designed & developed **framework-agnostic Web Components library** enabling seamless interoperability across Angular, React, and Vue ecosystems, while mentoring cross-functional teams on adoption, integration patterns, and best practices for implementation.
- ② Architected & delivered an end-to-end full-stack DevOps dashboard using Nuxt.js, integrating GitHub-based workflow for automated deployments, environment provisioning, traffic routing & PR management.

Ameex Technologies, Chennai | Front-end Developer | Oct'19 – May'22

- ② Delivered **30+ production-grade React applications** with modular component design and shared UI frameworks deployed across **4+ enterprise client projects**.
- ② Improved frontend performance metrics by **25%** through implementation of code splitting, route-based lazy loading, and optimized asset bundling, raising Lighthouse scores from **70 to 90+**.
- ② Integrated lightweight ML-driven recommendation systems into client-side applications, contributing to a **15% increase in user engagement** through early personalization features.
- ② Supported junior developers in adopting modern React practices and coding standards, improving sprint-level code review efficiency by **20%** and strengthening engineering consistency.
- ② Led **greenfield architecture and end-to-end frontend development** of an enterprise web platform, translating Figma designs into scalable, production-ready applications.
- ② Defined and implemented a comprehensive design system prior to development, ensuring consistent UI/UX standards, improved reusability, and accelerated delivery across the platform.

PERSONAL PROJECTS

YouRAG – YouTube RAG Assistant

- ② Architected an end-to-end RAG system for YouTube transcripts using Python, FastAPI, LangChain, LangGraph & Google ADK, implementing ingestion pipelines, chunking, embedding generation, and Chroma-based hybrid semantic search with agentic context-retrieval loops, dynamic prompt construction & tool-orchestrated LLM workflow to deliver sub-second, **context-aware responses across 100+ videos and reduce multi-video research time by 60%**.
- ② Built a scalable full-stack conversational platform with a high-concurrency backend and React-based chat UI featuring real-time LLM token streaming via SSE, message virtualization, markdown-based citation rendering, and persistent chat state management for seamless, low-latency user interaction.

Allergy Identification Progressive Web App (Multimodal Mobile AI)

- ② Developed cross-platform multimodal AI Progressive Web App capable of detecting allergens using smartphone camera image streams & free-text inputs, enabling sub-second classification on hardware.
- ② Implemented a hybrid inference pipeline combining TensorFlow.js for on-device real-time predictions & cloud-based LLM APIs for deeper contextual reasoning over detected ingredients & textual inputs and designed a offline-first PWA architecture featuring custom Service Workers, optimized caching strategies & responsive mobile-first UI design.

High-Performance JSON Visualizer (Developer Utility Tool)

- ② Built high-performance frontend developer tool for exploring, filtering & analyzing deeply nested JSON structures, optimized for large-scale corporate payloads **10K+ nodes**, reducing debugging effort by **30%**.
- ② Implemented advanced performance optimization techniques including list virtualization, React memorization strategies & rendering pipelines to ensure zero lag under heavy hierarchical data loads.

AWARDS

IBM Watson Hackathon Winner

- ② Won **3rd Prize at the IBM Watson Hackathon**, demonstrating strong problem-solving ability, innovation & hands-on technical expertise in building impactful solutions under competitive conditions.

Consistent Top Performer (2 Consecutive Years)

- ② Recognized as a **top performer for 2 years**, consistently delivering high-quality engineering outputs, improving team productivity, and contributing to successful project execution.

Company Stock Award Recognition

- ② Awarded **company stock grants for exceptional contributions**, reflecting sustained impact on enterprise projects and consistent delivery of high-value engineering solutions.

PERSONAL DETAILS

Address: Bangalore, India
Languages: English